

Financial dominance: why the ‘market maker of last resort’ is a bad idea and what to do about it

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Financial regulation, influenced by mainstream models of banks as sources of liquidity risk, rather than post-Keynesian models of banks as liquidity providers, fostered the growth of the repo market and shadow banking from the 1980s on. In the repo market banks provide funding liquidity, which interacts with market liquidity to cause liquidity spirals. Examples are the 2008 and 2020 crises. The central bank has responded to these crises by stepping in to purchase assets. This paper argues that we are now in a situation of ‘financial dominance’ where sophisticated financial market participants design financial instruments in expectation that the central bank will rescue the debt market from liquidity spirals. To address the problem of financial dominance, this paper argues for regulation based on the post-Keynesian approach to banks such that central bank financial support is provided via the banking system. This paper recommends (i) requiring that non-financial corporation repo and derivatives liabilities be supported by bank lines of credit and that banks face constraints on the ability to sell collateral into an illiquid market; (ii) requiring that monetary run-risk intermediaries, such as money market funds, be regulated as banks; and (iii) stricter regulation of BBB rated holdings for investment funds (or non-monetary run-risk intermediaries) when they are prohibited from holding below investment grade assets. These reforms must be accompanied by a clear policy that, in the event that any bank requires re-capitalization by the government, existing shareholder interests will be wiped out, and ownership will be transferred to the government.

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1. Introduction

In the mini-budget crisis of September 2022, the Bank of England was forced to rapidly implement large-scale purchases of gilts in order to restore order to a sovereign debt market disrupted by a sudden onset of massive sales. A pension fund investment strategy called ‘liability-driven investment’ (LDI) was quickly identified as the primary source of the destabilizing dynamics (Cunliffe, 2022, pp. 6–7). This was an example of financial dominance: the central bank was forced by financial market instability to intervene directly to support markets in a way that has non-neutral effects (Brunnermeier, 2016; Diessner and Lisi, 2020; Tooze, 2022b). Here, we emphasize that the destabilizing dynamics caused by LDI were predictable as was the central bank response, and that the result was that unsophisticated market participants were caught unprepared for the sudden fluctuation in gilt prices, while sophisticated traders anticipated both the price fluctuation and the central bank intervention (see Section 4).¹ This paper proposes regulatory reforms that are designed to forestall the need for such central bank interventions in markets by reintermediating the banking system in order to ensure that the central bank does not need to intervene directly in financial markets, but can support financial markets through the banking system.

The rationale for these proposals derives from the extensive post-Keynesian literature on the indispensable role played by banks in making modern economic activity possible by financing it. This literature on banks as liquidity providers contrasts with the mainstream approach to banks which frames banks as sources of liquidity risk. The paper argues that bank regulation over the past few decades has erred by relying on the faulty mainstream approach to banking. The resulting structural changes in debt markets leave them subject to sudden downward price spirals; since 2008 such price spirals have transformed the role of central banks into reactive buyers of last resort for financial markets. To escape this situation, central banks will have to revise the theoretical framework on which financial regulation is built. This paper provides a first attempt to address the problem of financial dominance in modern financial markets by proposing a regulatory framework that is constructed on the more realistic post-Keynesian model of banking.

There is a broad literature on financialization that is related to this paper and includes the study of non-financial corporations’ (NFCs’) growing reliance on ‘market-based’ finance, the shift in bank activities towards supporting financial market transactions, and the shift in profits towards the financial sector since the 1980s (Krippner, 2005; Lapavistas, 2011; van der Zwan, 2014; Christophers, 2018). Like Diessner and Lisi (2020) this paper focuses more narrowly on the role of central banks in the process of financialization, and in particular, on the consequences for the financial system of the current commitment to act as a ‘market maker of last resort’, but goes beyond a description of structural changes and sets forth policy proposals to address the situation. This paper is more optimistic than Sweeney (2019) about the possibility of re-intermediating the banks and proposes regulatory changes that would require that NFC repo/derivatives transactions be supported by bank credit lines and that money market funds and bank-financed finance companies operate under bank charters.

¹ This definition of financial dominance refers to a situation where the central bank is reactive and has little choice but to intervene to support markets. Another use of the term in the literature refers to a much milder form of financial market influence on central bank decision making: in this usage, financial dominance is when fear of triggering a financial crisis affects a central bank’s decision-making process with respect to monetary policy (Fraga *et al.*, 2003; Hannoun, 2012; Coeuré, 2015; Gabor, 2016).

The [next section](#) discusses banking theory, contrasting the mainstream view of banks as sources of liquidity risk with the post-Keynesian view of banks as liquidity providers. The [third section](#) describes the role of the central bank in crises and compares the two main approaches used to stabilize markets: the central bank ‘put’ and the banking approach, relying on the post-Keynesian framework to explain the advantages of the latter. [Section 4](#) discusses financial dominance in detail and how it interacts with the central bank ‘put’. Sections five and six consider how to eliminate the structural factors in our financial system that have forced the adoption of the central bank ‘put’ policy: [Section 5](#) provides a taxonomy of non-bank entities that cause financial instability, and [Section 6](#) provides recommendations for how to reinstate that banking approach using this taxonomy. [Section 7](#) concludes.

2. Banks: the economy’s liquidity providers

A vast literature frames the banking system as a source of liquidity risk and the central bank as a potential source of moral hazard when it provides liquidity insurance. A cornerstone model in this literature is [Diamond and Dybvig \(1983\)](#) where banks borrow short, for example, by issuing deposits, and lend long, e.g., mortgages.² This is a source of liquidity risk and instability: if too many borrowers withdraw their deposits the bank can run out of easy-to-sell, liquid assets and be forced to sell its long-term assets at fire sale prices and ultimately lose so much money that it cannot pay off all of its depositors.³ To eliminate the risk of bank runs, the central bank as lender of last resort can buy the assets at an above-market price— at the cost of creating incentives for banks to take on too much risk.⁴ This approach is echoed throughout a literature that emphasizes ‘the liquidity and credit risks inherent to the banking business’ ([Ugolini, 2017](#), p. 106). Even those authors who seek to explain the benefits of credit-based financial systems dwell on the problem of liquidity risks (e.g., [Bindseil, 2019](#), p. 32). This facilitates a framing of the central bank as the liquidity risk insurer for the banking system ([Bindseil, 2019](#), pp. 34–35; [Tucker, 2014](#), pp. 1, 11; [van t’Klooster, 2021](#), p. 52; see also [Volcker, 1984](#), p. 548).

While this paper acknowledges the value of this framework for understanding some aspects of banking, we argue that it overemphasizes the importance of the central bank and underemphasizes the liquidity *provision* aspect of private banking. Liquidity refers to the ability to access cash, or a means of payment that is acceptable to all. Because bank deposits are an accepted means of payment, a bank can typically ‘create’ liquidity by making a loan and increasing its deposits outstanding.⁵ As [Diamond and Dybvig \(1983\)](#) makes clear, a central bank that acts as lender of last resort such as the Federal

² Because the depositors in the Diamond and Dybvig model deliver goods to the bank that then distributes them to producers, a core post-Keynesian critique of the model is that it fails entirely to model financial/monetary instruments. Here, the model is discussed as it is in the mainstream literature, accepting the conflation of real and financial/monetary deposits in order to demonstrate the broader defects of the model.

Note that a more traditional approach to banking views banks as engaging in a limited form maturity transformation by lending on a short- or medium-term basis for commercial purposes and specifically not engaging to any significant degree in mortgage lending ([Sissoko, 2016](#)).

³ While the Diamond and Dybvig model posits benefits over autarky in the form of risk sharing, the benefits typically seem small relative to the risks that are being taken (see e.g. [Gu et al., 2020](#)).

⁴ Note that [Diamond and Dybvig \(1983\)](#) focus on deposit insurance as a solution to the problem of bank runs, but also observe that a lender of last resort can solve the problem in their model.

⁵ As [Unger \(2016, p. 5\)](#) points out ‘credit expansion in the traditional banking system is not subject to the laws of supply and demand to the same extent as it is for other parts of the financial system’. See also [McLeay et al., 2014](#); [Deutsche Bundesbank, 2017](#); [Cheng and Torregrossa, 2022](#).

Reserve or the Bank of England, is a solution to the problem of bank runs. Thus, banks that have access to central bank support are a source of liquidity for the economy. While it is true that a bank may be constrained from expanding liquidity by capital or liquidity regulations, most banks manage their balance sheets so that these constraints do not bind.

Such bank-based liquidity is however not the only form of liquidity. Asset owners have access to market liquidity when they choose to sell. Market liquidity may however not be reliable: e.g., when the announcement of the intention to sell an asset causes the price at which the asset can be sold to fall.⁶ Asset owners more recently may also have access to the repo market⁷ and to funding liquidity, when there is a lender who is willing to accept the asset as collateral and to provide a loan against a large portion of the asset's value. In practice, however, reliable access to funding liquidity can only be provided by banks (see below).

The emphasis on liquidity risk rather than the liquidity provision aspect of banking is natural in a world of poorly regulated banks with weak balance sheets including a great deal of liquidity risk, but it tends to obscure the fact that the foundations of modern banking lie in an environment with unlimited liability and very wealthy bankers who were extremely well capitalized, not least because their extensive estates served as a backup source of funds for creditors. In this environment losses due to liquidity events were borne almost entirely by the bankers themselves (Sissoko, 2018). Thus, there is another foundational model of banking where well-capitalized banks provide elasticity to the monetary system by making the loans that facilitate entrepreneurship and underlie economic growth (Schumpeter, 1939).⁸ In Schumpeter's framing the basis of creative destruction is banks' creation of new liquidity to fund new ideas, thereby allowing the old processes to continue to receive the trade finance that they have in the past, stabilizing relative prices temporarily, so that the underlying forces of demand and supply can be allowed to make the ultimate determination—through the price mechanism—of which products fail and which succeed. While there is strong historical evidence that the flourishing of such a banking system was only possible in the presence of a central bank that stood ready to support the private banks through liquidity crises (Sissoko, 2016, 2022a),⁹ the focus here is on the role played by the private banking sector in providing the liquidity that supports economic growth.¹⁰

Following Schumpeter there is an extensive literature on the essential role played by banks in a modern economy.¹¹ Schumpeter's student, Minsky (1980, p. 32), summarized pithily the underlying relationship between the financial and the real economy—or alternatively between banks and productive innovation—as 'a fundamental maxim of economics: only that which is financed can occur'. Chick and Dow (1988) use

⁶ This is modelled as the problem of selling 'long-term' assets in Diamond and Dybvig (1983).

⁷ 'Repo' is short for repurchase agreement, which is a collateralized loan that is structured as a sale and repurchase of the asset used as collateral.

⁸ Bindseil (2019, p. 175) emphasizes the importance of an elastic source of central bank money, but largely focuses on the period before the private banking system extended this elasticity in the way that Schumpeter describes. (Sissoko, 2021a) models the role of banks in supporting an elastic currency that can adapt to changing economic circumstances.

⁹ For a different point of view on this topic, see Selgin and White (1987).

¹⁰ Desan, 2015 and Hockett and Omarova, 2017 present a 'finance franchise' view of banks where the central bank is viewed as the ultimate source of liquidity, but the micro-level provision of liquidity—in the sense of the distribution of that liquidity across the economy—is delegated to the privately operated banks. See also Ingham, 2004.

¹¹ See Realfonzo, 1998 for an overview of this literature.

this framework to draw a close connection between modern banking systems—with interbank lending markets and central bank support—and the robustness of regional economic growth. Stock flow consistent models were developed for the purpose of integrating the analysis of the financial/monetary and real aspects of the economy, and they explicitly model the banking sector (Godley, 1999; p. 397; Godley and Lavoie, 2007; Dos Santos, 2006; p. 545; Caverzasi and Godin, 2013, p. 169). Another post-Keynesian approach to modelling the role of banks and the fact that financing precedes production in a modern economy is circuit theory (Graziani, 2003; Dos Santos, 2006; p. 550; Smithin, 2016). Rochon and Rossi (2007) integrate circuit theory into the broader Keynesian literature by articulating the various roles played by the central bank in supporting the bank-based monetary system. Arestis and Sawyer (2006) compare the mainstream and post-Keynesian approaches to the role of the banking system as an important transmission mechanism between central bank interest rate policy and economic activity and demonstrate that the post-Keynesian approach has important implications for monetary policy and prudential regulation.¹²

A critique of the post-Keynesian literature has been that the models are too ‘bank-based’ and fail to account for modern financial engineering (Taylor, 2008; Lavoie, 2019; Nersisyan and Dantas, 2017, pp. 297–298).¹³ This paper explicitly addresses the problems created by modern financial engineering—and proposes a solution based on a post-Keynesian framework. To be more specific, the financial plumbing that supports financial engineering is found in the repo/derivatives market, where financial instruments, like bonds and securitizations are used as collateral.¹⁴ The way this market functions—and does not function—has played an important role in each of the twenty-first century financial crises. The other components of financial engineering that have played a destabilizing role in crises are shadow banks, such as money market funds and commercial paper conduits, that rely on financial markets to borrow and lend.¹⁵ These non-bank financial intermediaries have the same risk of runs that the Diamond-Dybvig model depicts, but unlike banks do not have access to the last resort lending function of the central bank. Section 6 relies on a post-Keynesian approach to develop proposals for stabilizing both the repo/derivatives market and run-risk intermediaries.

The post-Keynesian ‘only that which is financed can occur’ approach views markets—and market pricing—as dependent on financing, and therefore dependent on the banks that historically provided this financing. In this framework banks are both liquidity providers

¹² Note that Kashyap *et al.* (2002) is a mainstream model that frames banks as liquidity providers. They however build on a three-period framework with a risk of liquidity demand in the middle period (like Diamond and Dybvig) and rely on exogenous demand for deposits and for loan commitments. Thus, they do not draw the connections between lending, deposit-creation and real economic activity drawn by Schumpeter and the post-Keynesians, but like Diamond and Dybvig choose not to model the value of bank liquidity provision. Instead, they acknowledge the isomorphism between deposits and loan commitments and argue that *because* banks hold liquid assets to honour deposits, this makes it less costly for them to also offer loan commitments (which may in theory be drawn down in different circumstances from deposits). Additional efforts to capture the role of banks using mainstream modelling techniques are found in Sissoko, 2007 and 2021a.

¹³ Responses to this critique include that banks continue to play a crucial role in the loan origination process (Unger, 2016; Lavoie 2019; Bouguelli, 2020) and that banks are skilled at circumventing the regulations that are designed to circumscribe their ability to originate and hold loans (Lavoie, 2012–13; Werner, 2016).

¹⁴ Since the mid-2000s these two markets, both of which rely heavily on financial instruments as collateral, have been closely integrated. For this reason, it is referred to as the repo/derivatives market here.

¹⁵ Commercial paper is a type of debt that is unsecured, issued by a business, and has a maturity of less than one year. A commercial paper conduit refers to a financial structure that funds itself using commercial paper, and that typically invests in other assets, such as auto loans or consumer credit.

and the underlying source of any ‘market liquidity’ as the banks provide the financing that makes purchases on the market possible. The mainstream approach to banks by contrast imagines markets that are not bank-dependent and treats ‘market liquidity’ as a fundamental property of markets themselves.¹⁶ Similarly, the mainstream literature on funding liquidity typically fails to identify the ‘financiers’ who provide the funding, even though, in practice, such funding is heavily bank dependent (see below). This mainstream misconception regarding the degree to which the various forms of liquidity are reliant on banks is what underlies the regulatory decisions that have created modern financial engineering (Sissoko, 2019; see also Sissoko, 2010, 2017). Market liquidity and funding liquidity are closely interrelated. Funding liquidity is dependent on the market value of the asset, which determines how much the lender is willing to lend. As a result, funding liquidity will become scarce precisely when markets become illiquid and prices fall. At the same time, in markets where reliance on funding liquidity is common, if funding liquidity becomes scarce, asset owners may be forced to reduce their holdings of assets. Sales of assets will reduce their market price and impact on market liquidity. Thus, Brunnermeier and Pedersen (2009) explain how the co-dependency of market liquidity and funding liquidity can create an adverse feedback loop that drives asset prices down creating a liquidity spiral.¹⁷ Such liquidity spirals are typically present in modern crises and are closely associated with the repo/derivatives market where funding takes place.

Funding liquidity is only available on the repo/derivatives market when the lender (or its agent) has control of the collateral and has the contractual right to sell it promptly in the event that the borrower defaults.¹⁸ The lender’s administration of collateral, however, requires not just a trading account, but also the ability to keep the various assets posted as collateral inappropriately segregated accounts: in short, it requires the skills of a bank.¹⁹ Thus, funding liquidity is heavily dependent on bank-based liquidity—and this is why post-Keynesian analysis is so relevant to the modern financial system.²⁰ In addition, when rating agencies set ratings for the ‘market-based’ instruments used by shadow banks they demand bank liquidity commitments for these instruments, on the basis that banks are the entities with ‘ready access to funds’ and central bank liquidity support, and so are ‘ideal providers of liquidity’ (Moody’s, 2003, p. 43; see also Stigum and Crescenzi, 2007, pp. 977, 995). Furthermore, as noted above because of the close connection between funding liquidity and market liquidity, for assets that are widely used as collateral for funding on repo markets, bank liquidity underpins not only funding liquidity, but also market liquidity and market pricing.

The modern repo/derivatives market—and the accessibility of funding liquidity to entities such as hedge funds—is a relatively new phenomenon starting in the 1980s

¹⁶ Cochrane (2014) for example argues that shifting from bank finance of business activity to equity-funding would eliminate runs and thereby have a stabilizing effect on financial markets. In short, some mainstream economists conflate the assumptions of their modelling frameworks (e.g. perfectly frictionless markets) with the implications of the models, and as a result, imagine that real-world markets will behave like the markets in their models. Empirical evidence rebutting Cochrane (2014)’s approach includes Choi and Kronlund (2017), Quinn and Turner (2020) and Breckenfelder *et al.* (2021).

¹⁷ The same interrelationship can work in both directions and thus can also drive prices up.

¹⁸ The ‘safe harbour’ amendments to the US Bankruptcy Code of 1984, 1990, and 2005 made it legally enforceable to include a clause in the contract allowing repo and derivatives collateral to be sold even if the borrower declared bankruptcy (Sissoko, 2010).

¹⁹ Indeed, the tri-party repo market developed precisely because investment funds wanted to lend on the repo market but did not have the capacity to manage the collateral (Garbade, 2006).

²⁰ Indeed it should come as no surprise that the repo/derivatives market, where this funding takes place, grew to its current importance precisely because it was backstopped by a too-big-to-fail bank (Sissoko, 2020).

and growing dramatically in the 2000s.²¹ Thus, the extent to which funding liquidity is made available by banks is a recent phenomenon. It is for this reason that neither banks nor central banks were historically expected to play a major role in supporting the value of privately issued financial market instruments: in the absence of widespread access to funding liquidity, liquidity spirals were rare.

How did the growth of the repo/derivatives market happen? In the intellectual environment described above where the mainstream Diamond-Dybvig approach is generally accepted and banks are viewed as a source of liquidity risk rather than as a source of liquidity provision, regulation has heavily favoured markets over banks—implicitly assuming that the neo-classical economic model of ‘markets’ is accurate and that markets are not subject to liquidity events. The result of this approach has been, first, the growth of a repo/derivatives market that was viewed as using collateral and market liquidity to protect banks from liquidity risk (Sissoko, 2010, 2017), second, the growth of non-bank financial intermediation—shadow banks—together with ‘market-based’ lending, and, ultimately, crises that demonstrate conclusively that so-called market-based lending is a source of financial instability.²² Commercial paper markets were shown to be unstable in 2007 (Sissoko, 2013). Money market funds have repeatedly required support from the state or central bank (BIS, 2020). The same is true of the repo market (Gabor, 2016; Sissoko, 2020). Indeed, a much broader list of unstable shadow banks can be drawn up than there is space for here. The underlying problem has been a failure to understand what it is that banks do, why they are important, and the mechanism by which they can play a role in stabilizing financial markets.

In this environment where regulation fosters market-based financial instability, since 2008 we have repeatedly seen central banks step in as buyers of last resort for financial markets (discussed in detail in section 3). This role is sometimes called ‘market maker of last resort’ and has defenders as a natural evolution of central bank liquidity provision (Mehrling, 2011).²³

The post-Keynesian approach to the role of bank liquidity in the economy provides a framework for critiquing this evolution as destabilizing. After all, an important implication of the post-Keynesian approach is this: because the modern economy runs on bank-based credit, price levels are indeterminate, and bank liquidity (which includes the combination of central bank + banking system liquidity) profoundly influences both (i) the general level of prices—which we typically view as guided by central bank policy—and (ii) depending on how bank liquidity flows, relative prices too. Indeed, it is precisely because central banks are aware of the potential for affecting relative prices that they frequently treat ‘neutral’ implementation of monetary policy as a goal (see van t’Klooster, 2021). Central banks—and financial regulators more generally—have, however, failed to take into account that the banking system is also an important provider of liquidity and that it is very important to monitor whether the banking system

²¹ From 1984 on, the passage of ‘safe harbours’ for repos and derivatives that ensured that the collateral could be sold even after the debtor had declared bankruptcy played an important role in this development (Sissoko, 2010), as did the merger of JPMorgan and Chase Manhattan banks in 2000 (Sissoko, 2020).

²² In addition to the ‘safe harbours’ in bankruptcy, Basel II was another part of the misguided regulatory structure that promoted (i) collateralized lending by banks over unsecured lending and therefore the growth of the repo/derivatives market, (ii) shadow banking vehicles like securitization (since credit ratings determined capital requirements and securitization could convert a bundle of risky loans into a package 50% of the value of which was rated as ‘safe’), and (iii) the use of derivatives that could be used to reduce capital requirements on a loan to zero.

²³ Others both recognize the important role that the ‘buyer of last resort’ plays and the serious implications this has for financial market incentives (Hauser, 2021).

itself is providing liquidity in a sufficiently neutral manner and not favouring, for example, housing markets, or certain well-connected hedge and private equity funds.

Another aspect of the fact that prices are a function of the flows of bank liquidity, is that sovereign debt does not have a meaningful ‘exogenous’ market price. In practice, in most countries, sovereign debt is a core banking system asset, and furthermore, most modern high-income countries have a history of relying on central bank and/or banking system support of the value of sovereign debt when the state’s fiscal needs are high. While the price of sovereign debt in these countries can still carry important information as it changes from day to day or year to year, that value is typically also determined by deep institutional factors that affect the broader role of sovereign debt in the financial system.

For example, throughout the centuries during which the UK’s fiscal capacity developed, it relied heavily on the Bank of England both for short-term funding and to support the issue of long-term debt²⁴ (Sissoko, 2021b, 2022b). The cooperation of the Federal Reserve with the US Treasury during the wars of the twentieth century is well-established (Meltzer, 2003). More recently central banks and fiscal authorities cooperated during the COVID-19 crisis (Ryan-Collins *et al.*, 2023). In short, what is meant by ‘the market value of sovereign debt’ is very complicated, precisely because the interrelationship of the central bank and banking system with that debt means that frequently sovereign debt is not properly characterized as having an exogenous ‘market’ value.²⁵ Regulatory reforms post-2008 have meant that sovereign debt is now the most important source of collateral in the repo/derivatives market.

To summarize the post-Keynesian framework that models banks as essential liquidity providers to the modern economy is necessary:

- First, to provide a counterbalancing view to the mainstream approaches that treat banks and markets as separate phenomena instead of recognizing that markets do not function independently of bank financing.
- Second, to guide our understanding of financial regulation. Because post-Keynesians frame banks as liquidity providers instead of as a source of liquidity risk, the regulatory focus shifts to asking how (and where) banks are providing liquidity. Because post-Keynesians view markets as relying on bank-provided liquidity, regulation should never draw on so-called ‘market liquidity’ to support banks.
- Third, to make clear how dramatically the modern financial structure has changed with the growth of repo/derivatives markets that provide access to bank-based funding liquidity that was not available historically and that has transformed and destabilized market liquidity.

²⁴ By providing short-term loans to the syndicate that won the bid for the debt and then distributed it to final purchasers (Sissoko, 2022b). In particularly difficult circumstances these short-term loans could be rolled over by the Bank as a form of support for the money market.

²⁵ From this perspective the case of the European Central Bank (ECB) is distinct from that of central banks historically, because the ECB is a monetary authority that interacts with multiple fiscal authorities and because it was not founded with a clear remit of supporting the value of the various sovereign debts with which it interacts. Indeed, the European debt crisis was driven by the ECB’s collateral policy, which—at the time—relied on ‘markets’ to determine the value of the various European sovereign debt issues (see van t’Klooster, 2021; Orphanides, 2022). Only when ECB President Mario Draghi adopted the policy of ‘whatever it takes’ did the ECB implicitly acknowledge the inherent endogeneity of central bank policy to the valuation of sovereign debt. More recently the ECB’s Transmission Protection Instrument is designed to give the ECB the policy space to act as a traditional central bank by supporting the value of sovereign debt when appropriate—and importantly when not in conflict with broader European Union policy goals (ECB, 2022). This may be comparable to the traditional Treasury-central bank cooperation that frequently arises during periods of high debt issuance in countries with a simpler monetary system.

3. Stabilizing unstable money markets: the advantages of the banking approach

Just as Minsky argued that thwarting mechanisms were needed to stabilize the economy and financial markets (Dafermos *et al.*, 2023), this paper proposes that, when financial markets work well, it is because they have been stabilised. Historically, stabilisation has taken one of two forms: the banking approach, where the central bank uses banks as an intermediary to stabilize markets, and the direct approach or central bank ‘put’ where the central bank acts directly on markets. It is the latter that generates a risk of financial dominance. While the banking approach was more important through most of the twentieth century, the ‘put’ is a simpler mechanism and will be described first.

The central bank ‘put’ refers to the central bank’s capacity to support asset prices by buying the assets outright without any obligation for a private sector entity to compensate the central bank for losses on the assets.²⁶ A dramatic recent example of the central bank ‘put’ is the Federal Reserve’s purchases of US Treasury securities in March and April 2020 (see Figure 1). Over the last two weeks of March, the Federal Reserve purchased nearly 5% of the total stock of marketable Treasuries outstanding, and almost 8% of the total stock of Treasury bonds outstanding (Sissoko, 2020). This is a clear

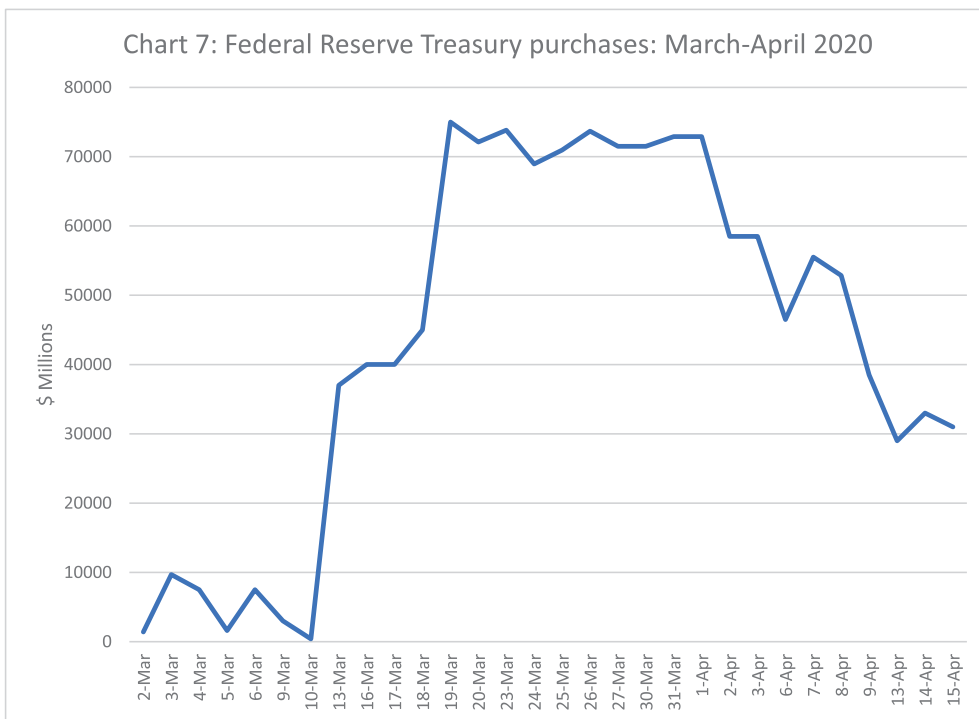


Figure 1. Federal Reserve’s purchases of US Treasury securities in March and April 2020

Source: Federal Reserve Bank of New York Permanent Open Market Operations data
(Federal Reserve Bank of New York, 2020)

²⁶ Note that obligations to compensate the central bank can take many legal forms: endorsement on a negotiable instrument such as a commercial bill (e.g. nineteenth-century discount window), holding a junior lien on an asset where the central bank holds a senior lien (e.g., the Fed’s Maiden Lane conduits), a recourse loan from the central bank (e.g., modern discount window), etc.

example of the central bank's use of its purchasing capacity to put a floor on an asset market. While this is the most dramatic example of the 'put', the Federal Reserve's purchases in 2020 were not limited to Treasuries—the 'put' was also used to support corporate debt markets more generally through programs such as the Primary Market Corporate Credit Facility and Secondary Market Corporate Credit Facility (Cavallino and De Fiore, 2020). A more recent example was discussed in the introduction: the Bank of England purchased Gilts at the end of September 2022 (Bank of England, 2022).

As was demonstrated in both spring 2020 and autumn 2022, the central bank 'put' is an effective means of supporting asset prices when markets are unstable and prices are collapsing—at least for those countries that are able to do so without seeing a significant decline in the value of their currency in domestic or international markets. The 'put' however raises concerns about financial dominance that are discussed in section 4.

The banking approach is an alternative stabilizing mechanism for financial markets. The models for the banking approach are the British banking system in the nineteenth and twentieth centuries and the US banking system in the twentieth century after the reforms of the 1930s.

There is a crucial similarity between the 'put' mechanism and the banking approach: it is the central bank that is the ultimate source of liquidity. The key difference is that in the 'put' scenario the central bank balance sheet is directly at risk, whereas in the banking scenario the central bank balance sheet is protected by the balance sheets of the banks. This structure is essential to the banking approach because of the healthy incentives that it creates for financial markets and for financial market pricing—and specifically *not* because there is any concern about the central bank's capacity to bear losses.

The banking approach has the following structure: the central bank provides ample liquidity against private sector assets, but only when they are posted as collateral by a bank against a recourse loan, or the equivalent.²⁷ In a recourse loan, if the underlying collateral goes bad, the bank still owes the full value of the loan to the central bank and so it is the bank that takes the loss, not the central bank. Of course, if the borrowing bank goes bankrupt, it is possible that the central bank takes a loss on the loan—but this is precisely what is meant when one says that the central bank is protected by the balance sheet of the bank.

From the earliest days of banking the recourse loan structure was the norm. Modern banking evolved from the bill of exchange that financed European trade throughout the early modern era, and the holder of a bill could only use it to make payment by guaranteeing the payment of the bill himself. By the nineteenth-century central bank lending took place in the bill market (Sissoko, 2022a; see also Bindseil, 2019, p. 111), and this practice of requiring the borrower to be liable for the full amount of the loan continued when central banks began to make collateralized loans similar to modern discount window loans in the nineteenth century (Sissoko and Ishizu, 2025). When the Federal Reserve was founded in 1913, it imitated European central bank lending norms and also required that all loans be structured as recourse loans. Thus, in March

²⁷ The claim here is not that there is only one legal structure for the banking approach, but rather that it is always the case that in the event that there are losses on the private sector asset, the bank will have a legal obligation to compensate the central bank for those losses.

2008 when the Federal Reserve provided a non-recourse loan to induce JPMorgan Chase Bank to purchase Bear Stearns, it was described as ‘unprecedented’ (Dodd, 2008; see also [Congressional Oversight Panel, 2009](#), p. 116; [Swagel, 2009](#), pp. 24, 42).²⁸

What was different in 2008 that led to this unprecedented central bank action? JPMorgan Chase (JPMC) provided Bear Stearns with prime broker services, which meant that in the repo/derivatives market, JPMC was the principal bank that handled Bear Stearns's collateral and lent against it. In short, JPMC provided funding liquidity to Bear Stearns, and it was JPMC that had evaluated Bear's collateral portfolio and found it inadequate, forcing Bear to close. The Federal Reserve acted because the liquidity spiral that would have resulted from Bear's bankruptcy and the liquidation of the collateral that Bear had posted against its derivatives/repo liabilities would have destabilized the financial system ([Geithner, 2008](#), pp. 114–15; [Cox, 2008](#), pp. 191–92).²⁹

The failure to understand the post-Keynesian approach to banking underlies the regulatory decisions that created the modern repo/derivatives market that has so dramatically destabilized modern financial markets. The current solution, which is equally founded on a failure to understand the post-Keynesian approach to banking, has been the central bank ‘put’. In [section 6](#) this paper provides an alternate proposal for restructuring funding liquidity so that the banking approach can once again play a stabilizing role in the financial system.

Before focusing on the problems created by the central bank ‘put’ it is useful to review the broader role of the central bank in its traditional, historical role. The operation of a traditional banking system (e.g., as conceived by [Schumpeter, 1939](#)) works well only if the banks are good at underwriting and originating loans that will be paid back, and, furthermore, even when the banks are doing their jobs well this system is subject to crises of confidence that may be triggered by real events such as a natural disaster or war. It is in this second case that we see the necessity of a central bank to support the system. The natural disaster or war has an effect on asset prices and on the value of the banks' assets. If one bank fails to impose losses on the other banks, that can set off a chain reaction of additional failures as the banking system and the monetary system are destabilized. This is where the central bank steps in as lender of last resort, to give the banking system time to determine the true value of the losses and their allocation. On the one hand, central bank liquidity support via recourse loans ensures that banks can carry assets that are only subject to temporary price dislocations without taking a loss, while at the same time requiring the bank to ultimately recognize the losses if the price change is not temporary, but instead reflects underlying fundamentals. On the other hand, this support also avoids a sudden chain of failures and allows those banks that have losses that will force them to close, to fail not at the outbreak of the news of the disaster, but only after its implications have been fully understood. In short, when a crisis of confidence is at risk of causing well-managed banks to fail, the central bank in its traditional lender of last resort role uses recourse loans to stop the doom loop by lending to banks and supporting the value of their assets.

²⁸ As [Swagel \(2009, pp. 24, 46–47\)](#) explains non-recourse lending became an important element of the ‘rescue’ toolkit at the Fed after the Bear Stearns failure.

²⁹ Note that bankruptcy imposes an automatic stay on creditors that would prevent the sale of this collateral, if not for ‘safe harbour’ amendments to the Bankruptcy Code that were passed in the US starting in 1984—on the basis that these ‘safe harbours’ were necessary to stabilize the financial system! ([Sissoko, 2010](#)). Such legal changes are the principal reason that the repo/derivatives market is so important today.

That the banks must be good at underwriting and originating loans is also very important: as [Schumpeter \(1939\)](#) pointed out, when bank origination practices are flawed, the economy will be subject not just to crises of confidence, but also to solvency crises—and the latter can lead to depression. This then is the purpose of requiring the banks to insure the central bank against losses on their assets: the alignment of incentives is the best way to ensure that banks will do a sound job of lending and originating assets that promote healthy economic activity.

Of course, even in an environment where incentives are aligned, things are not so simple. Human judgment is imperfect. Since the early nineteenth century banking systems have faced the problem that ‘innovative’ banks—which are actually just underpricing risk—can gain market share. If the innovative bank survives for a few years, competitor banks will watch their market shares fall, and inevitably someone in the competing bank will argue that the only solution is to adopt the ‘innovative’ approach since it has (short-term) demonstrated success (cf. [Smith, 1936](#), p. 88). In this situation, it may be the case that many other banks follow the first bank’s ‘innovation’. In this scenario, even though incentives are properly aligned, aggressive competition in the banking sector can result in a general problem of debt that underprices risk—with a crisis as the ultimate result ([Sissoko, 2016](#), p. 6; [Goodhart, 1988](#), p. 48. See also [Akerlof and Romer, 1993](#), p. 41).

It is for this reason that banks are regulated and central bank support is never guaranteed, but always subject to be withdrawn if the central bank deems the prospective borrowing bank to be insolvent.³⁰ This practice of central bank supervision of the banking system provides both additional protection for the central bank when supporting the banking system through a crisis and also the opportunity in normal times to monitor bank lending practices and take preventive action when there are signs that a bank is underpricing risk.

In short, the advantage of the traditional banking system over the ‘put’ as a means of providing central bank liquidity is precisely that regulated banks sit between the central bank and any losses. This structure enables the central bank to protect the financial system pro-actively by regulation³¹ in normal times, and also ensures that incentives are aligned even when the central bank is engaged in crisis lending—the bank that holds a bad asset will ultimately bear any losses on it.

This then gives two important advantages of the banking approach over the central bank ‘put’:

- The incentive structure that is created for financial market participants. Whereas the ‘put’ gives financial market participants the opportunity to transfer losses to the central bank, the banking approach ensures that private sector losses are borne by the private sector. In short, the moral hazard of the central bank’s liquidity provision is minimized by this structure (cf. [Schnabel, 2020](#), p. 5).
- A smaller information burden for the central bankers in crisis. If the collateral supporting a central bank crisis loan is overvalued, it doesn’t really matter under the banking approach, because any losses will still sit with the borrowing bank. On the other hand, the ‘put’ will require evaluation by the central bank of the ‘fundamental’

³⁰ The distinction between solvency and illiquidity is rarely clear. Thus, this is a situation where the central bank must exercise judgment. Indeed, it was understood in the early years of central banking that the primary determinant of solvency for a bank was the willingness of the central bank to support the bank ([Baring, 1967](#), p. 22).

³¹ Regulation here includes both macro- and micro-prudential regulation.

value of assets when they purchase them—and no mechanism for recovering losses due to overvaluation. This latter problem is a concern principally due to the distortionary effect that it can have on private sector decision making.

Overall, the banking approach is in many ways an ingenious way of providing liquidity to financial markets and of stabilizing value on money markets, while at the same time limiting the distortions created by the central bank's support. That said, the banking approach can only work if the central bank is in the dominant position, and no 'too big to fail' bank can effectively hold a gun to the economy's head and demand special treatment from the central bank. In short, the banking approach can only be adopted together with a framework in which poorly managed banks are closed down and resolved with the costs borne by the bank owners and employees. The banking approach also needs to be adapted to address modern derivatives/repo markets, as is discussed in [section 6](#).

4. Financial dominance: why the central bank 'put' must be avoided

While the preferability of the banking approach when the asset facing a liquidity event is privately issued may be clear, some have argued that repeated use of the central bank 'put' to support the value of sovereign debt is not a matter for concern (see e.g. [Davies, 2022](#)). Arguably this approach is naïve. When it comes to sovereign debt, the greatest concern with the central bank 'put' is that it is a reactive tool—the central bank is literally forced by debt market dynamics to act. Adam [Tooze \(2022b\)](#) explains the issue well: 'What is at stake is not fiscal dominance - the central bank following the lead of the elected government - but financial dominance. The central bank is being forced to act by financial market stress'.

This is a very dangerous place for a central bank to be because financial markets are extremely good at exploiting these supportive tools: when central bank intervention is a sure thing, then financial instruments that are designed to profit from that central bank intervention will be designed. Furthermore, when the central bank intervenes the instrument is likely—by design—to benefit the designers at the expense of other parties, including third parties affected by financial market volatility. Thus, 'financial dominance' refers not just to a situation where the central bank is forced by financial market instability to intervene directly to support markets, but also to a situation where the central bank typically ends up favouring some market participants over others.

The Liability Driven Investment (LDI) funds that forced Bank of England intervention in September 2022 comprise an example of a financial instrument that was designed to force the Bank of England to act. LDI funds are offered by major asset managers³² and sold as an investment to smaller UK pension funds. [Cunliffe \(2022, pp. 6–7\)](#) reports that these pooled LDI funds were a particular focus of the Bank's concerns and indicates that in the likely event that these funds defaulted it was the anticipated sale of the gilts these funds had posted as collateral³³—and the broader implications of such sales into an already extremely illiquid market—that was the immediate catalyst for the Bank's actions. As the collateral demands of LDI strategies have been clearly understood by financial professionals for at least a decade ([Wood, 2022](#)), one

³² The three asset managers providing these funds are Blackrock, Legal & General, and Schroders.

³³ Unsurprisingly, the repo/derivatives market, where collateral is posted, is once more the locus of financial instability.

can only conclude that the LDI outcome that was realized in September 2022 was one of the potential outcomes that was considered when asset managers, such as Blackrock, designed the funds. Indeed, Blackrock's statements after the crisis indicate that there was an expectation that the Bank of England would intervene. Mark Weidman, a 'top Blackrock executive', stated 'This is a market with a very small number of natural buyers called pension funds and the Bank of England, so when they have margin calls and there's no one on the other side of the trade that creates a little bit of instability' (Agnew and Masters, 2022). The inclusion of the Bank of England as a 'natural' buyer of gilts in a period when its stated policy was one of selling gilts is remarkable—and speaks directly to the expectations that Blackrock held when it designed the fund.

The question needs to be asked whether we now live in a world of financial dominance, and whether the 2008 financial crisis was just the first crisis of financial dominance—one that would be followed by ever more frequent episodes of financial dominance.

Recall that one of Alan Greenspan's legacies was the 'Greenspan put', as he was the first central banker to take the position that financial institutions had no obligation to protect themselves against certain types of risk. One of Greenspan's earliest actions as Fed Chairman was to preside over a New York Fed coordinated bailout of US investment banks that were at risk due to the 1987 stock market crash—in an era before Fed lending to investment banks was authorized by statute. And Greenspan (1998, p. 167) stated explicitly in 1998 that regulators should not

'require individual banks to hold capital in amounts sufficient to fully protect against those rare systemic events which, in any event, may render standard probability evaluation moot. The management of systemic risk is properly the job of the central banks. Individual banks should not be required to hold capital against the possibility of overall financial breakdown. Indeed, central banks, by their existence, appropriately offer a form of catastrophe insurance to banks against such events.'

So-called 'market-based finance' has arguably grown up in an effort to profit from this promise of central bank intervention. The repo/derivatives market grew in importance after Greenspan's promise of insurance for the financial system against systemic risk. As the liquidity spirals and fire sales associated with margin call dynamics in collateralized funding markets are precisely the sort of liquidity events that fall under the heading of systemic risk, there was an implicit expectation that when these liquidity dynamics were observed, it would be a problem, not for the individual financial institutions, but for the central bank.³⁴ And so, in 2008 the Federal Reserve was forced to act to address catastrophic liquidity dynamics in privately issued assets, not just by providing non-recourse loans, but also by creating the Primary Dealer Credit Facility, the Term Securities Lending Facility, by granting exemption letters to commercial banks with dealer bank affiliates, and by engaging in quantitative easing designed to support the US mortgage market.

While regulators after 2008 were committed to the reform of the financial system, they also began to embrace the idea that central banks had a role to play as 'market

³⁴ In 2012 the International Capital Markets Association (2012) put it very clearly—virtually restating what Greenspan had said in 1998: 'the authorities can be expected to intervene as lenders of last resort to ensure the liquidity of the system as a whole. For their part, market users should be expected to remain creditworthy and to have liquidity buffers sufficient to sustain themselves until official intervention restores sufficient liquidity to obviate the need for fire sales'.

maker of last resort’—which in practice is a euphemism for ‘prop trader of last resort’ (Sissoko, 2019) and the central bank ‘put’. In the United States, an important regulatory goal was to reform repo and derivatives markets so that the principal collateral was sovereign debt—which was viewed as highly liquid and much less like to require ‘prop trader of last resort’ intervention. After all, when it comes to sovereign debt the ‘prop trader of last resort’ role is related to the traditional role played by central banks in sovereign debt markets which is discussed in Section 2 above. Even so, there is an important distinction to be drawn between the traditional coordination between the Treasury and the central bank in times of heavy debt issuance—where the risk is that monetary policymaking is dominated by fiscal policymakers (i.e., fiscal dominance)—and the ‘prop trader of last resort’ which is a role that is made necessary by the actions of financial market participants, not Treasury. Here, the risk is financial dominance—or in other words, monetary policy-makers risk losing their independence to financial market participants. This should be viewed as a much more serious risk than the loss of independence to fiscal authorities in a country with a democratically elected government.

In short, the problem with central banks embracing the ‘prop trader of last resort’ role in sovereign debt markets is that they thereby incentivize sophisticated financial market participants to create products that rely on central bank intervention as a back-stop. This is likely to have the effect of ensuring that the sophisticated financial market participants do not lose money, while less sophisticated participants are likely to face losses due to the price volatility that always occurs during these episodes. The problem with the central bank ‘put’—or alternatively the ‘prop trader of last resort’—in sovereign debt markets is that by ceding to the exigencies of financial dominance, central banks will lose any ability that they may have had in the past to implement policy in a neutral manner.

The problem then becomes how to eliminate the need for the central bank to step in reactively as ‘market maker of last resort’. The answer is that we need to eliminate the structural factors in our financial system that have created liquidity spirals and forced the central bank to react. This is in fact easier than it may at first appear.

5. Non-bank financial intermediaries (NBFIs) and the banking approach to stabilization

Even after convincing the reader that the central bank ‘put’ is dangerous policy, the problem remains: what can be done to transform a financial environment that is currently entirely reliant on the central bank ‘put’ to function?

It is helpful to start by looking in detail at one of the chains of events that amplified the recent COVID-19 crisis. Pension and insurance funds use derivatives to hedge interest rates and foreign exchange risk. As a result, the initial March 2020 dislocation in sovereign debt prices meant they faced margin calls. To meet these calls, they drew down money market fund (MMF) holdings, which forced the MMFs to sell their holdings which happened to be heavily concentrated in Treasuries and repos of Treasuries (Schnabel, 2020, p. 3; Setser, 2020). The ultimate result was that an initial dislocation in sovereign debt markets led inevitably to more sales of sovereign debt and a worsening dislocation of those markets—and more margin calls—and finally a liquidity spiral.

This example illustrates how the underlying problem with this ‘market-based system’ was the assumption that sovereign debt markets would be immune to liquidity events: sovereign debt was used as ‘safe’ collateral, MMFs offered instantaneous liquidity against their holdings of ‘safe’ assets, and sovereign debt markets were somehow expected to be robust to massive, coordinated sales by some of the biggest entities in the financial system. In short, the March 2020 crisis demonstrated that in the repo/derivatives market, the negative feedback loop between market liquidity (sales) and funding liquidity (reduced financing for assets) applies to sovereign debt as well as to privately issued debt. The idea that the sovereign debt market would be more stable was an illusion. Thus, the central bank was forced to step in as a reliable source of liquidity.

The challenge then is to re-design financial regulation so that it reflects the post-Keynesian framework, recognizes the essential importance of bank-based liquidity (and the inadequacy of market and funding liquidity), and shifts central bank support of the financial system to the banking approach rather than using a direct ‘put’.

The first step is to create a taxonomy of non-bank entities that are susceptible to generating liquidity risk in financial markets. Note that this is a larger category than that of non-bank financial intermediaries (NBFIs). To determine this, two questions need to be asked:

1. Can the owners of the financial assets issued by the entity run? In other words, does this entity issue an asset that, like a bank, promises immediate liquidity and/or access to funds? These are **run risk intermediaries**.³⁵
2. Can the entity experience a run on collateralized obligations or assets? Given our current market structure this is equivalent to the question: does the entity use derivatives or repo or a comparable form of collateralized debt? These are **derivatives and repo market participants**.³⁶

The second question may capture entities that do not typically fall under the heading of a non-bank financial intermediary and yet due to the current structure of derivatives markets and collateralized lending markets, may still play a role in financial instability. The second question will be discussed first.

5.1 Derivatives and repo market participants.

Since the mid-2000s derivatives and repo markets have been reformed so that collateral can flow smoothly from one market to the other and so that the legal framework for making margin calls and seizing and selling the collateral (or margin) is both robust and rapid (Sissoko, 2010). For this reason, margin calls in derivatives markets and margin calls in repo markets will tend to have comparable effects. Furthermore, these markets have grown dramatically in size since the 2008 crisis when unsecured money markets first faltered and then fell into disuse and when central banks demonstrated that they stood ready to backstop this collateralized system (*ibid.*). These collateralized markets are constructed upon the faulty theory that financial markets themselves can be a source of liquidity (Sissoko, 2019). As was discussed in section 2 above, in practice ‘market’ liquidity is heavily dependent on the availability of financing from

³⁵ Cf. Ricks (2016). This concept is related to Acharya and Oncu (2013)’s systemically important assets (SIAs).

³⁶ These participants are using what Acharya and Oncu (2013) call systemically important liabilities (SILs).

banks. In fact, we have seen repeatedly that margin calls work as a coordination device draining liquidity from financial markets (Sissoko, 2020). Today these markets have become themselves a source of instability.

More recently the world has seen the risk generated by this collateralized system on commodity markets when a single very large derivatives trader triggered a dislocation on the nickel market that forced the London Mercantile Exchange not just to suspend trading, but to void more than an hour of trades (Mackintosh, 2022). In modern markets, market dislocation can be caused by any large participant in derivatives or repo markets, including producers of commodities who seek only to hedge their price risk over time (Tooze, 2022a). Furthermore, even those who *lend* on repo markets can experience a run if borrowers decide that the entity is no longer a reliable counterparty (Geithner, 2008, p. 114; Duffie, 2010, pp. 23–42).

The fact that these markets have the effect of making all large market participants concerned for market stability is a clear indicator that these markets are fundamentally flawed and need root and branch reform. This will be discussed below.

Figure 2 depicts the taxonomy of entities that create liquidity risk. The figure indicates that derivative and repo market participants is a category that includes all of the other entities. This reflects that derivatives and repo are used ubiquitously in modern financial markets and that virtually every significant financial market participant is likely to also make use of these collateralized markets. In short, while it is certainly possible to imagine a run-risk intermediary that does not use repo or derivatives, in practice it is very difficult to come up with an example.

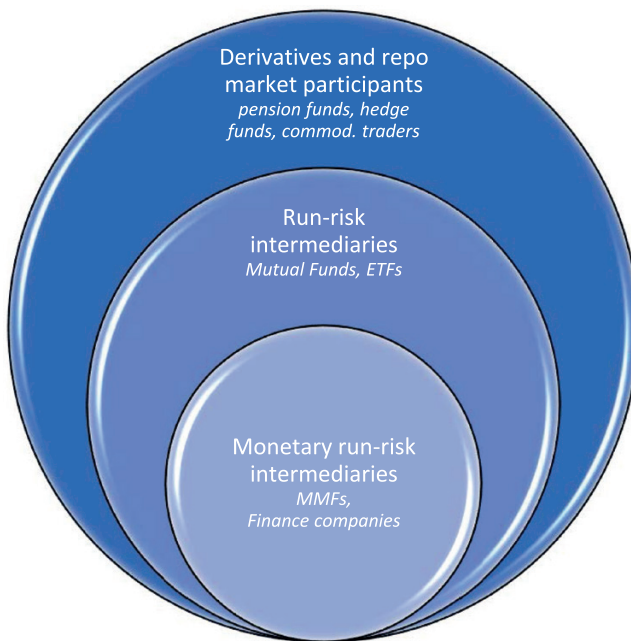


Fig. 2. Taxonomy of non-bank entities that create liquidity risk
Source: Author.

5.2 Run risk intermediaries.

The run risk intermediaries are divided into two categories: those that play a role in expanding the money supply, *monetary run risk intermediaries*, and those that do not, *non-monetary run risk intermediaries*.

5.3 Monetary run risk intermediaries

play a role in expanding the money supply—and increasing the capacity of the banking system to make loans—by either (i) providing short-term wholesale funding for banks or (ii) relying on short-term wholesale funding from banks to engage in lending activities or (iii) issuing a near-deposit liability and either directly funding commercial business activity or doing so supported by off-balance-sheet bank liabilities. The different types of monetary run-risk intermediaries are depicted using balance sheets in Figures 3–5.

The first category, the provision of short-term wholesale funding for banks, focuses on NBFIs, such as money market funds, that support banks’ lending capacity by providing banks with an alternate source of funding for deposits. Thus, even though from a consumer perspective MMFs ‘compete’ with deposits, in practice when the consumer shifts her deposits into MMFs, the MMFs in turn provide funding to the banks. This insertion of the NBFI into the bank’s funding system is depicted in Figure 3 using the example of a money market fund (MMF).

Without NBFI	Bank		Consumer	
	Assets	Liabilities	Assets	Liabilities
	Loans	Deposits	Deposits	
With NBFI	Bank		MMF	Consumer
	Assets	Liabilities	Assets	Liabilities
	Loans	Bank CP	Bank CP	MMF shares
				MMF shares

Fig. 3. NBFIs that provide wholesale funding for banks, e.g., commercial paper, and thus intermediate deposits.

Source: Author.

Without NBFI	Small business		Bank	
	Assets	Liabilities	Assets	Liabilities
		Loans	Loans	Deposits
With NBFI	Small business		Finance company	Bank
	Assets	Liabilities	Assets	Liabilities
		Loans	Loans	Repo
				Repo
				Deposits

Fig. 4. NBFIs, such as finance companies, make loans relying on wholesale funding from banks and thus intermediate loans.

Source: Author.

Corporation		MMF	
Assets	Liabilities	Assets	Liabilities
	CP*	CP*	MMF share

* Bank provides off-balance-sheet credit and liquidity support so that the CP has a credit rating high enough for the MMF to hold it.

Fig. 5. NBFIs, such as MMFs, rely on off-balance-sheet bank commitments to lend and issue near deposits liabilities.

Source: Author.

The second category is NBFIs which increase banks' capacity to make loans by holding the loans (which under the Basel accords may have high capital requirements if held by the bank) and using repo to borrow from the bank (which will have lower capital requirements for the bank because it is collateralized). The insertion of the NBFi into the bank lending process is depicted in Fig. 4, using the example of a finance company.

The third category refers to NBFIs that like money market funds expand banks' capacity to lend because the loan is not held on the bank's balance sheet, but instead is supported by an off-balance-sheet bank liability. These entities may appear to have a balance sheet that is very similar to a bank balance sheet, but in fact, rely on off-balance-sheet bank support to lend. This case is depicted in Figure 5, using the example of an MMF.

Overall, money market funds fall into the monetary run-risk intermediary category for both reasons (i) and (iii): a core investment for prime money market funds is bank-issued commercial paper, and they also invest in non-bank commercial paper that is supported by bank credit and liquidity facilities (Sissoko, 2024). Finance companies, which typically are funded by bank-issued warehouse lines of credit (Jiang *et al.*, 2020), often fall under the second heading. (Please note that if banks were to be included in this taxonomy, they would be monetary-run risk intermediaries.)

Other entities may also fall into this category—though most of the possibilities that come to mind existed prior to the 2007–09 financial crisis. Structured investment vehicles (SIVs) such as Sigma Finance were funded by commercial paper and medium-term notes while providing wholesale funding to banks (Mollenkamp, 2008). In 2006 a late cycle collateralized debt obligation (CDO) product was also developed that was funded by asset backed commercial paper (ABCP), supported by off-balance-sheet bank liquidity 'puts', while investing in mortgage-backed assets. Bear Stearns' Klio Funding products are examples (Henry and Goldstein, 2007; Financial Crisis Inquiry Commission, 2011, p. 136ff). It is worth noting that SIVs and CDOs that issued ABCPs also participated in derivatives and repo markets thus they were subject to runs not just by the holders of the commercial paper that they issued, but also by their collateralized creditors.³⁷ Thanks to the robust regulatory focus on the problems

³⁷ Observe that the role played by banks in supporting the growth of vehicles such as SIVs and ABCP-funded CDOs falls squarely under the well-established phenomenon of adverse competitive dynamics in banking that were discussed in section 3. After one poorly managed bank sponsors the creation of a vehicle that underprices the risk the bank faces, the bank necessarily gains market share in comparison with banks that price the risk correctly. If regulators do not put a stop to this behaviour promptly, then the other banks will see their market share decline. The 'market' signal is that the other banks should also underprice risk. Indeed, this is the best explanation of the notorious comment by Chuck Prince, Citigroup CEO in 2007: 'When the music stops, in terms of liquidity, things will be complicated. But as long as the music is playing, you've got to get up and dance. We're still dancing' (Nakamoto and Wighton, 2007). Adverse competitive dynamics preclude reliance on market signals in banking and make bank regulation a challenging task that requires minute, detailed knowledge of the banking industry.

generated by these monetary-run risk intermediaries, it is difficult to find more recent examples of such vehicles – aside of course from money market funds and finance companies.

5.4 Non-monetary-run risk intermediaries

include investment funds that allow investors to sell out of their holdings at any time, such as mutual funds and exchange-traded funds. When the funds face an unusual flow of redemptions, they are unlikely to hold enough liquid assets to meet the redemptions and will be forced to sell the underlying assets. Thus, the selling pressure is transmitted directly onto financial markets.

Non-monetary run risk intermediaries do not just generate financial stability risk because they can be forced to sell assets, but also because they may participate in repo and derivatives markets. Exchange-traded funds often rely on derivatives to manage their interests in underlying assets, and in the USA, even mutual funds may be subject to ‘hidden’ leverage when they use derivatives to increase their exposure to the underlying assets (Jacobson, 2009).³⁸

Given this vocabulary let us reconsider the discussion above of the March 2020 sovereign bond incident: the liquidity spiral was due to the interaction of large derivatives market participants (pension and insurance funds) with monetary run risk intermediaries (MMFs). This analytic structure helps to explain why entities such as pension funds, that are not inherently subject to run risk, can play an important role in financial instability: the problem is the structure of derivatives and repo markets.

6. Recommendations

This section explores how we need to transform the modern financial system in order to re-establish the banking approach to central bank liquidity provision.

6.1 Repo and derivatives markets

The current structure of repo and derivatives markets was adopted on the basis that it was a way to protect the financial system, and banks in particular, from a chain of failures (Sissoko, 2010). However, the system of protecting banks from losses by relying on repo-type collateral contracts does not work. This has been evident since the 2008 crisis. The reason the system doesn’t work is intuitive: when the defaulting borrower is large, the amount of selling that needs to take place in order for collateral to fully protect the lenders is so large that there are not enough buyers on the market to support the vast influx of sales orders. Central bank interventions, including the Fed’s Primary Dealer Credit Facility and the Term Securities Lending Facility, were necessary to prevent a liquidity-driven collapse in asset prices (Sissoko, 2010). Just 12 years later the Fed once again had to step in with purchases that would support asset prices (BIS, 2020).

The question then is: how should repo and derivatives markets be reformed so that they are no longer protected by a central bank ‘put’, but instead by the banking approach? That is, how do we bring the risk of lending to repo and derivatives market

³⁸ On the US regulation of mutual fund derivatives exposures see International Monetary Fund (2015, p. 23) and Securities and Exchange Commission Rule 18f-4.

participants onto the books of the banks, where any support from the central bank for collateral markets can be provided through the banks as intermediaries?

The answer is that non-bank derivative and repo liabilities should be supported by bank lines of credit. A bank cannot play the role of an agent liquidating a position for a client but must be ready to become the principal on the loan. The banks may, of course, demand collateral to support those lines of credit; however, some criteria need to be set forth for the conditions under which the banks may not sell the collateral they have obtained by foreclosure, worsening a liquidity crisis. When the banks are precluded from selling the collateral, they will have available the alternative of posting the collateral in a recourse loan to the central bank in exchange for liquidity.

Given the importance of banks to repo/derivatives market lending today, this is in some sense a minor change—it would simply impose the requirement that *all* non-bank repo and derivatives liabilities be supported by a bank line of credit. The significant changes would be (i) in the contractual status of the collateral securing a repo or derivatives liability, and (ii) in agency markets like the tri-party repo market. In the first case, the bank making the loan would no longer have the contractual right to sell the collateral in all market conditions, as is the case today. When market conditions are difficult, the bank would be required to hold onto the asset and bear the risk that it permanently falls in value. Carrying the asset would be supported by the central bank acting as a traditional lender of last resort—providing the bank with recourse loans. In the second case of agency markets like the tri-party repo market, all loans would be supported by a bank line of credit. Thus, if a liquidity event prevents the sale of the collateral and renewal of the repo loan, the bank credit line would be drawn down to honour the repo loan, and the bank would be in the same situation as if it had made the repo loan itself.

The goal of this policy is to stabilize prices on asset markets by putting an end to collateral-driven liquidity spirals. The liquidity spiral is stopped because funding liquidity is only provided by banks and these banks cannot sell the collateral into an illiquid market.³⁹ Banks when they provide credit lines to support repo and derivatives will have to look to the longer-term value of the collateral posted and will bear some risk that the collateral falls in value before it can be sold. In short, to the degree that banks offer funding liquidity, they will bear some of the risks of the assets that they fund. This is good. Banks should be well-placed to evaluate the quality of the collateral posted and to protect themselves by doing this evaluation. Their need to engage in this process will help to ensure that prices on financial markets are robust and reflect the underlying fundamentals of the assets.

To understand why this policy change is important, consider the recent dislocation in the nickel market that forced the London Mercantile Exchange to take the unprecedented action of cancelling more than an hour's trades (Burton and Farchy, 2022a). The over-the-counter derivatives exposures of some of the world's largest banks played a very important role in this episode (Farchy and Burton, 2022b). Had the banks been precluded from selling the collateral into an illiquid market, it is all but certain the dislocation on the market would have been much smaller.

Indeed, it is unclear why a producer that can indisputably deliver the underlying contract at the appropriate date should have to post cash or bonds to cover its nominal

³⁹ Note that there is no need for a bank to stop a liquidity spiral by making asset purchases in the same way that the central bank does. This is because the dynamics of the liquidity spiral are stopped at the funding liquidity stage.

liability during a liquidity event. It would make sense for collateral calls to be limited to those who are on the buy side of the contract or who are not hedging real positions. Such distinctions could easily become part of bank management of credit lines supporting derivatives transactions in place of the current collateral regime.

6.2 Monetary run risk intermediaries

These intermediaries should be incorporated into the banking system and the bank regulatory system (cf. [Ricks, 2016](#)). Money market funds, like the repo market, are founded upon a misconception about market liquidity: while US regulation requires them to invest in high-quality short-term assets, in certain financial conditions market liquidity is inadequate, and for this reason, they were protected by the central bank ‘put’ in both 2008 and 2020.⁴⁰ In 2008, US money market funds offered redemptions at a fixed value (like bank deposits) despite the risk that under severe market conditions the promised redemptions would not be feasible forcing the liquidation of the fund. The 2008 failure of the Reserve Primary Fund forced the US Treasury to temporarily guarantee US money market fund holdings, even though this had not been anticipated and no insurance premiums had been paid, and in addition, the Federal Reserve created a Money Market Mutual Fund Liquidity Facility (MMLF) to support the value of money fund assets. In 2020 institutional money market funds no longer offered fixed-value redemptions—so investors were notified upfront that the value of their investment could fall. This structure, however, may have aggravated the run dynamics in March 2020 when the Federal Reserve once again opened the MMLF to avert fire sales of money fund assets ([Eren *et al.*, 2020](#)).

Money market funds are fundamentally inconsistent with a banking approach to central bank liquidity. Under these circumstances, it makes far more sense to subject money market funds to bank regulation, than to try to fix a regulatory structure that has already failed twice in the past 20 years.⁴¹ Bank regulation includes the requirement of obtaining a bank charter, of paying into the deposit insurance fund, of being subject to the bank regulatory regime including the Basel Accords and the production of regular public reports, and of being subject to more detailed non-public supervision by bank regulators.

As was noted above, one of the important advantages of the banking approach to central bank liquidity provision is the alignment of incentives in financial markets. Both captive finance companies and those that are funded by a consortium of banks should be viewed as a form of regulatory arbitrage: in some cases, the corporate form is being used to socialize losses that may arise from poor underwriting practices ([Pistor, 2019](#), p. 53), and in others, the goal may be to benefit from the finance company being subject to different regulations than the bank. It is hard to see how there is any social gain from this complicated regulatory structure. Indeed, research has shown that these ‘non-bank’ lenders serve a less creditworthy client base and also when financial

⁴⁰ The Fed’s programmes in 2008 included the Asset-Backed Commercial Paper Money Market Mutual Fund Liquidity Facility, the Commercial Paper Funding Facility and the Money Market Investor Funding Facility. In 2020 the Fed’s programmes included a new Commercial Paper Funding Facility and a Money Market Mutual Fund Liquidity Facility.

⁴¹ Note that this does not preclude mutual funds from offering short-term bond funds. What it does preclude is a special regulatory category for money market funds including specific limits on the fund’s investments. Any short-term bond funds would fall under the standard mutual fund regulatory regime—which would force investors to bear the responsibility of monitoring whether the fund was investing consistent with its prospectus, and the option of suing in the event that the fund was not doing so.

conditions become stressed act pro-cyclically, withdrawing credit from these clients more quickly than banks (Aldasoro *et al.*, 2022). If the banks want to fund a lending activity, then they should be required to bring it in-house. Overall, any companies that are funded predominantly by wholesale bank lines of credit and either make loans (other than repo) or purchase financial assets should be treated as bank-regulated activities and incorporated into the banking system.

6.3 Non-monetary run risk intermediaries

The preceding reforms to the financial system will require any non-monetary run risk intermediaries who trade derivatives or borrow on repo to rely on bank credit lines to cover their liabilities—in the event of an adverse market and failure to meet a margin call, the bank lender may foreclose on any collateral that has been posted, but the bank will not be able to sell that collateral into an illiquid market. This is a banking approach to preventing liquidity spirals. Of course, it is possible that banks will find the prospect of relying on collateral under these conditions too risky and will fund a much smaller range of repo and derivatives activity than has been the norm in the recent past. This should be viewed as a healthy change in the markets, as risks that had previously been put to the central bank are being internalized with a natural effect on market activity. In short, the repo and derivatives markets reforms should be adequate to prevent investment fund withdrawals from being transformed into a liquidity spiral.

In addition, there will be no money market funds that are forced to sell assets due to investment fund withdrawals. Instead, such withdrawals may create stress for a bank. This situation can however be handled by central bank liquidity using the banking approach and by bank regulators.

While these reforms do a great deal to limit the risks generated by non-monetary run risk intermediaries, based on the behaviour of bond markets in March 2020, it is likely that some additional regulation of non-monetary run risk intermediaries may also be needed in order to forestall any need for a central bank ‘put’. In particular, corporate bonds outstanding are not just at unprecedented levels, but they are also of particularly low quality with half just one step above junk. If these low-quality bonds are downgraded then many investment funds will be forced to sell them: insurance companies, pension funds, and certain investment-grade funds will not be allowed to hold these bonds after a downgrade due to the requirements of regulation and/or the fund’s prospectus (Vissing-Jorgensen, 2020). As this situation very likely influenced the Federal Reserve’s decision to support corporate bonds in 2020, there is a clear need for regulation that puts an end to this form of downgrade risk. In particular, the holdings of insurance companies and pension funds of all tiers of BBB bonds⁴² should be limited by regulation to reduce the risk of massive forced sales—and at the same time there should be some very limited ability for a fund to hold a sub-investment grade position while in the process of seeking to sell it at a good price. Investment-grade bond mutual fund prospectuses should be drafted with similar limited ability for a fund to hold a sub-investment grade position while in the process of seeking to sell it at a good price.

Most important there must be a clearly stated policy that there will be no central bank ‘put’ supporting the corporate bond market. If ultimately, public pension

⁴² Investment grade bonds are rated AAA, AA+, AA, AA-, A+, A, A-, BBB+, BBB, and BBB-, where AAA is the highest ranking and BBB- is the lowest investment grade ranking. Thus, there are three tiers of BBB bond rankings. (Note that Moody’s uses slightly different indicators.)

funds and insurance companies need to be bailed out to preserve the well-being of the public, that is a matter for the fiscal authorities, not the central bank.

The purpose of these reforms and of the re-establishment of the banking approach to central bank liquidity is to ensure that the private sector bears the losses on financial assets and thereby to ensure that incentives are aligned and that distortions of financial asset prices are minimized. This of course means that any losses incurred by banks need to be imposed on the owners of the banks. In order for the latter to be the case, too big-to-fail banking must be addressed. Here the recommendation is that these reforms must be accompanied by a clear policy that in the event that any bank requires re-capitalization by the government, existing shareholder interests will be wiped out, and ownership will be transferred to the government. Banks cannot be exempt from the standard rules of capitalism.

We can contrast this banking approach to central bank liquidity provision with other proposals for stabilizing the financial system. Acharya and Oncu (2013), for example, advocate a focus on the regulation of systemically important assets and liabilities (SIALs) and propose in particular a repo resolution authority to address the risks created by the repo market. Writing in 2013, Acharya and Oncu assume that sovereigns like Treasuries and Gilts are not subject to the same repo dynamics that we witnessed with mortgage-backed securities in 2008. More recently the events of March 2020 and September 2022 have demonstrated conclusively that ‘safe assets’ are not immune to repo-driven liquidity spirals. The repo resolution authority that Acharya and Oncu propose for risky assets is not easily mapped onto ‘safe assets’—for example, the concept of a conservative valuation becomes harder to define when assets are not viewed as inherently risky. In addition, this repo resolution proposal is largely triggered by default and does not address clearly the problem that fire sale dynamics arise just as much (and perhaps even more) from repo borrowers avoiding default as from the sale of foreclosed collateral. The proposal here seeks instead to arrest the liquidity spiral at the source by preventing the bank holders of collateral from selling it into an illiquid market, while at the same time providing these banks with access to the liquidity that they need in the short run by application directly to the central bank for a recourse loan.

The proposal in Ricks (2016) is more closely related to this proposal because it too seeks to stabilize financial markets by reinforcing the principles underlying traditional bank regulation. In particular, Ricks proposes that only entities with a bank charter be allowed to issue short-term ‘run-prone’ debt⁴³ and that a core function of bank regulation is to police the boundaries of the banking system to actively prevent the growth of ‘unauthorized banking’. He envisions a world with a monetary authority, governing banks and the size of the money supply, but where a central bank is not strictly necessary even though it might be helpful for fine-tuning the money supply (Ricks, 2016, pp. 229–30).

Thus, the proposal here is much more modest than the Ricks (2016) proposal: here central banks will continue to play the role they play today as the ultimate source of liquidity for the economy; in addition, whereas Ricks would prohibit non-banks from borrowing on a short-term basis through for example the repo market (*ibid.*, p. 243),

⁴³ There is a trade credit exception to this rule (Ricks, 2016, p. 244). Given the development of the modern banking system out of the eighteenth-century system of trade credit in the form of bills of exchange (Sissoko, 2016), one can imagine that this exception could quickly grow extremely large—and would test the boundaries between ‘trade’ credit and financing more generally.

here banks would facilitate such borrowing for non-banks by providing credit lines to support collateralized borrowing both in repo and derivatives markets.

We acknowledge that the significant shift away from ‘market-based’ lending proposed here may in a crisis result in the expansion of bank balance sheets as credit lines are drawn down. As [Cooperman *et al.* \(2022\)](#) observe this may be costly for banks if their credit lines are not tied to a reference rate, such as the erstwhile LIBOR, that is sensitive to credit risk. We agree that our proposal may need to be supported by the development of a better substitute for LIBOR than the rates currently being adopted.

A more fundamental concern that this proposal for a banking approach to central bank liquidity provision raises is that by re-centring banks at the core of the financial system, financial instability will just shift from markets to the banks—and that massive too-big-to-fail banks will result. This criticism, however, underestimates the potential effectiveness of bank regulation. The sources of banking instability are well understood. Detailed data on bank financial positions is collected continuously and is monitored carefully by bank regulators. If there is one area where pre-emptive measures to avoid financial instability can be successful, it is within the banking system.

At the same time, it is true that the importance of maintaining a population of banks that is small enough to avoid ‘too big to fail’ should receive even greater emphasis amongst bank regulators than it already does (as many have argued before me, e.g. [Johnson and Kwak, 2010](#)). Many policies could be used to support this approach including, e.g., a cap on the size of a bank balance sheet, or a tax on large banks that grows over time to a prohibitive amount. In short, there is no reason whatsoever that the proposals in this paper should be permitted by bank regulators to be associated with the growth of the largest banks. This would only occur if regulators chose to forego their obligation to prevent this anti-competitive outcome.

7. Conclusion

Since the 1980s both shadow banking and the repo/derivatives market—along with access to the funding liquidity that the repo market provides—have grown dramatically. While this was viewed as ‘market-based’ finance, this growth was in fact heavily reliant on the banking system. And then in 2008, when the outcome of this massive increase in funding liquidity resulted in a liquidity spiral, the central banks had to step in as ‘market makers of last resort to prevent financial instability. By 2022 financial market participants had begun to rely on this new role of the central banks, as demonstrated by the liquidity spiral in Gilts that was set off by collateral calls on LDI funds. Now ‘financial dominance’ is the best descriptor of the current situation of central banks, where they are forced to react to the liquidity crises predictably generated by the private sector.

This paper builds on the post-Keynesian approach where banks are modelled as liquidity providers. Using this framework the paper advocates addressing financial dominance by adopting regulatory changes that move risk back into the banking system, where it can be properly monitored and easily supported by the central bank without distorting private sector incentives: when derivatives and repo liabilities are supported by bank credit lines and banks are prevented from selling collateral into an illiquid market, the central bank can act as a traditional lender of last resort by providing a recourse loan to a bank in need so that it can carry the collateral asset until the market for the asset stabilizes. Regulatory arbitrage is addressed by requiring certain shadow

banks to be regulated as banks. The most important advantage of this banking approach to central bank liquidity provision is that losses on private sector loans are borne by the private sector and not the central bank. The purpose of keeping private sector losses off of the central bank balance sheet is to protect the process of price formation on financial markets and avoid distortions.

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